Faculty of Computing

University of Sri Jayewardenepura

Undergraduate Final Year Project -Proposal Form

Please complete all of the following sections.			
Candidate Details			
Student Full Name:	Weragalage Don Yasiru Rasuwan Kalhara		
Registration No.	FC110596		
Examination No.	FC212023		
Supervisor Details			
Supervisor (1) Name	Dr. S.P.B.M. Senadheera	Department	Department of Information Systems Engineering and Informatics
Supervisor (2) Name	Mr. B.N.N. Rodrigo	Department	Department of Information Systems Engineering and Informatics
Project Details			
Title	**Sarusara AI**: An End-to-End System for Paddy Nutrient Deficiency Detection and Area-Based Fertilizer Recommendation Using Leaf Image Analysis		
Objectives	1) To build a data repository containing paddy leaf images representing Nitrogen (N), Phosphorus (P), and Potassium (K) deficiencies under varying field conditions. 2) To develop a mobile optimized object detection model to identify the diseased portions of the leaf image 3)To develop a multi-dimensional rule-based decision matrix that map Severity Index and the environmental factors 4) To develop an automated Area-Scale Translation Engine that converts raw nutrient deficit metrics into exact, localized mass weights of straight commercial fertilizers (Urea, DAP, and MOP). 5) To generate a system report to assess the performance of the deployed system.		
Project proposal	Refer Appendix II for project proposal guidelines. Please attach the Project Proposal to this form in the time of submission.		
Second Section (Only accessed by the Supervisors)			
Filled by Supervisor			
Decision	• Approved without changes • Approved with minor or major revisions • Rejected, requiring a new submission		
Comment			

Appendix I

Sarusara AI: An End-to-End System for Paddy Nutrient Deficiency Detection and Area-Based Fertilizer Recommendation Using Leaf Image Analysis

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Final Year Project Proposal

W.D.Y.R. Kalhara

FC110596

FC212023

2026

Appendix II

Content

1) Title page (Please refer Appendix I)
2) Introduction
3) Problem Statement
4) Aim/ Objectives (Around 5 objectives)
5) Method/s
6) Research Contribution
7) Project Timeline (Please refer the example project timeline (Appendix II)
8) References (LNCS Referencing style)
9) Certification section

Word Count: maximum 3000 including tables and figures

Introduction

Rice is the main staple food and is an important pillar of food security and economic stability in Sri Lanka. Around 1.8 million people (more than 25% of the overall domestic labour force) are actively involved in agricultural activities, of which about 16% is cultivated as paddy and 7% in terms of agricultural GDP. Although it is a cultural and financial significance crop, domestic paddy production is vulnerable to major biological and environmental risks. The estimated reduction in rice production due to biotic rice plant diseases caused by complex configuration of fungal, viral and bacterial pathogens is 37% in both global and local scale per annum. [3]

Yala season	Gross harvested extent		Average yield		Production	
	'000 acres	'000 hectares	Bushels per net acre	Kg per net hectare	Million bushels	'000 MT
2016	939	380	79.21	4,084	72.72	1,517
2017	584	236	75.54	3,895	43.58	909
2018	897	363	84.9	4,377	73.47	1,533
2019	855	346	84.68	4,366	72.82	1,519
2020	1,114	451	88.28	4,552	92.22	1,924
2021	1,228	497	83.57	4,309	100.08	2,088
2022	1,187	480	62.21	3,207	70.05	1,462
2023	1,215	492	74.12	3,822	87.1	1,817

Average yields are estimated as means of districts average yields

Figure 1: Gross Harvested Area, Average Yield and Production of Paddy 2016 – 2023 Yala Seasons (source: https://www.statistics.gov.lk/Agriculture/StaticalInformation/PaddyStatistics) [4]

Soils that are deficient in nutrients are a general impediment to yield, as well as direct pathogens. Visual expression of the underlying plant nutrient deficiencies is often confused by smallholder farmers in Sri Lanka because both are morphologically similar and extremely common. Add to this diagnostic error is the critical need for localized nutrient management, which is based on traditional nutrient management systems developed by the Department of Agriculture (DOA) which are based on standard soil analysis laboratories, field experiments, and nutrient adsorption studies. [1]

Routine soil-testing programs face structural constraints, however: they are long in the turnaround of logistics and labor intensive, with soil samples physically transported to remote testing centers; they routinely miss important local soil dynamics like adsorption capacity, which can result in inconsistent or delayed recommendations on the field. Therefore, farmers return to traditional "blanket" chemical fertilizer methods of application. It is an unquantified and non-site-specific management that is likely to cause critical nutrient imbalances, increase production costs and poor management of the surrounding groundwater table due to heavy chemical run-off.

Nutrient Constraint	Statistical Prevalence in Sri Lankan Paddy Fields	Key Visual Symptoms & Physiological "Diseases"
Nitrogen (N)	Universally fluctuating depending on seasonal top-dressing cycles.	General chlorosis (yellowing) starts from leaf tips, progressing inward, and reducing vegetative biomass.
Phosphorus (P)	95% of tested fields lack optimum levels; 44% fall below the critical threshold.	Stunted growth, dark green/purple leaves, massive lateral root curling, and sharp reduction in tillers.
Potassium (K)	77% of tested fields are deficient in plant-available Potassium.	Dark brown spots on older leaf margins, weak stems, and hyper-susceptibility to *Sesamia* pests and fungal diseases.
Zinc (Zn)	Prevalent in poorly drained wet zones and calcareous dry zones.	"Khaira" disease (rust-colored brown blotches, stunted seedlings, and brittle leaves).

Figure 2 Nutrient Deficiencies and Statistical Prevalence in Sri Lankan Paddy Fields)

Problem Statement

In Sri Lanka, smallholder paddy growers often confuse nutrient deficiencies with biotic diseases, and this leads to poor soil management practices and overall reduced yields. Current agricultural AI applications lack specificity when it comes to disease classification and cannot identify nutrient deficiencies or other key environmental factors (such as soil baselines and weather telemetry), which often manifest as additional diseases. Thus, no end-to-end system is available that can convert visual stress found in leaves to specific, commercial fertilizer recommendations (Urea, DAP and MOP) that are exactly proportional to the physical size of a farmer's field.

Aim and Objectives

The primary aim of this research is to design, develop, and evaluate **Sarusara AI**, an end-to-end mobile computing system that diagnoses precise multi-occurring nutrient deficiencies in paddy leaves through advanced computer vision and generates field-scaled, quantified commercial fertilizer recommendations using real-time contextual data fusion.

1) To collect, clean, and preprocess a localized dataset of at least 1,500 high-resolution paddy leaf images representing Nitrogen (N), Phosphorus (P), and Potassium (K) deficiencies under varying field conditions by August 2026.
2) To develop and train a mobile-optimized object detection model based on the YOLO architecture by October 2026, achieving a target Mean Average Precision of at least 90% in identifying multi-occurring nutrient deficiencies.
3) To design and implement a multi-dimensional rule-based decision matrix by November 2026 that dynamically maps the model's computed leaf Severity Index alongside regional soil baselines, crop phenology, and live weather telemetry APIs.
4) To program and validate an automated Area-Scale Translation Engine by December 2026 that converts raw nutrient deficit metrics into exact, localized mass weights of straight commercial fertilizers (Urea, DAP, and MOP) tailored to specific user field dimensions.
5) To generate a system report to evaluate the performance of the deployed system.

Methods

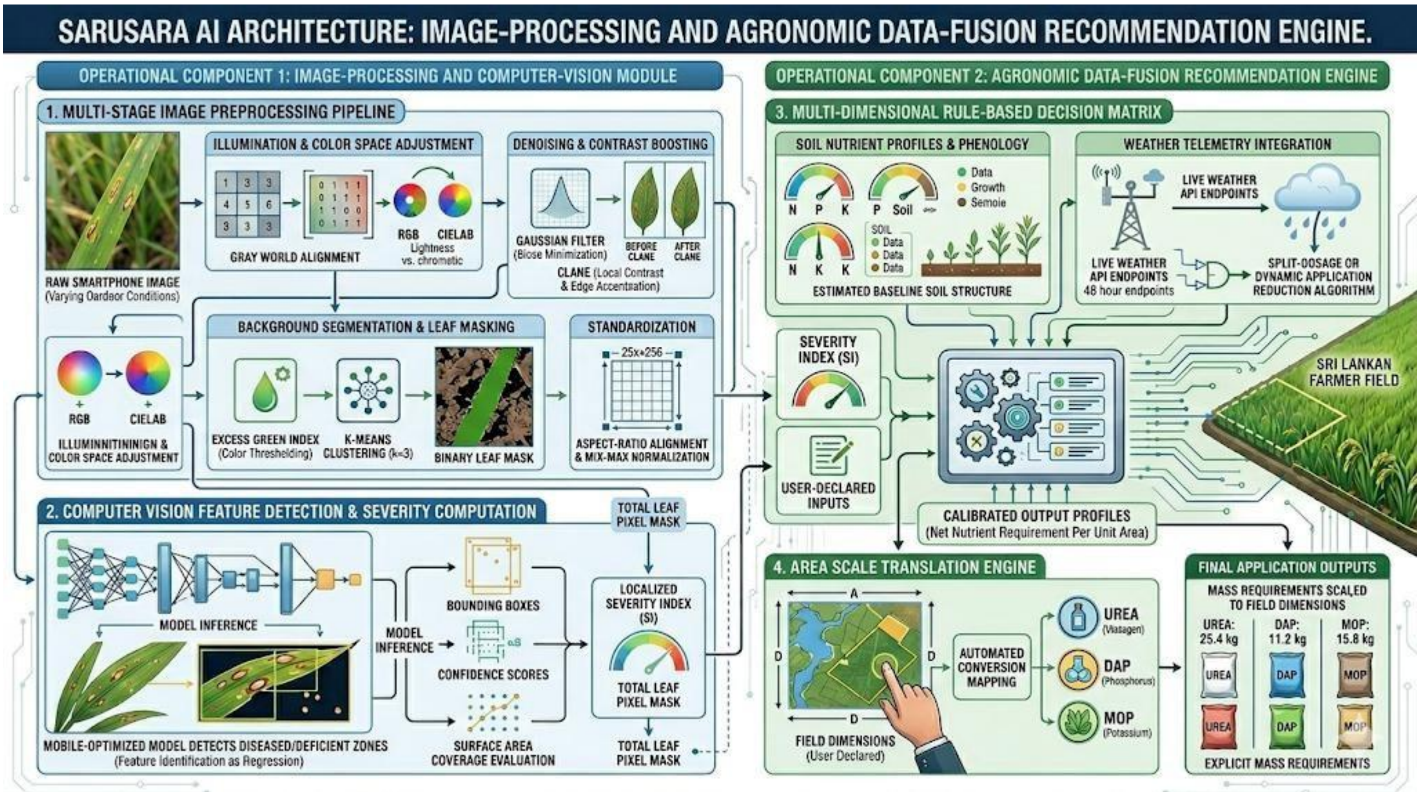


Figure 3 Architecture of Sarusara AI (Source: AI Generated Image With Gemini)

The architecture of Sarusara AI is designed across two interconnected operational components: an image-processing and computer-vision module, and an agronomic data-fusion recommendation engine.

Multi-Stage Image Preprocessing Pipeline

Raw images captured via standard smartphones under varying outdoor conditions undergo a structured pipeline to ensure semantic standardization:

- **Illumination & Color Space Adjustment:** Images are subjected to Gray World Alignment to mitigate color cast anomalies from ambient solar variances, followed by an RGB to CIELAB color space transformation to isolate lightness and chromatic components cleanly.
- **Denoising & Contrast Boosting:** High-frequency environmental noise is minimized using a Gaussian Filter kernel. Localized contrast enhancement is subsequently applied via Contrast Limited Adaptive Histogram Equalization (CLAHE) over localized tiling grids to accent high-frequency leaf lesion edges.
- **Background Segmentation & Leaf Masking:** To eliminate confounding soil and weed background pixels, color thresholding is enforced using the Excess Green Index. This is paired with K-Means Clustering (k=3) to compute a binary leaf mask, preserving only true leaf surfaces for down-stream machine learning analysis.

- **Standardization:** Final preprocessed elements are uniformly resized via aspect-ratio alignment to a 256x256 matrix and min-max normalized.

Computer Vision Feature Detection & Severity Computation

The standardized image is routed into a mobile-optimized YOLO object detection model. The choice of YOLO is guided by its capability to treat feature identification as a single regression problem, allowing it to isolate multiple co-occurring deficiencies on a single leaf asset simultaneously.
Upon running inference, the model generates specific bounding boxes accompanied by Class IDs and confidence scores. The application dynamically evaluates the surface area coverage of these bounding boxes against the total isolated leaf pixel mask area to derive a localized Severity Index (SI).

Computer Vision Feature Detection & Severity Computation

The computed Severity Index is fed into a multi-dimensional rule-based decision matrix alongside user-declared environmental inputs

- **Soil Nutrient Profiles & Phenology:** The system matches the computer-vision crop stress indicators with estimated baseline Nitrogen, Phosphorus, and Potassium (N,P,K) soil structures alongside current crop phenology (crop age and growth stage).
- **Weather Telemetry Integration:** The framework pulls local live weather telemetry via API endpoints. If heavy precipitation thresholds are exceeded within a 48-hour projection window, a split- dosage or dynamic application reduction algorithm triggers automatically to protect against fertilizer leaching.

Area Scale Translation Engine

The calibrated output profiles define the actual net nutrient requirement per unit area. The system runs an automated conversion mapping these requirements directly against commercial fertilizer formulations widely accessible to Sri Lankan farmers: Urea (for Nitrogen), Diammonium Phosphate (DAP, for Phosphorus), and Muriate of Potash (MOP, for Potassium). The final application outputs explicit mass requirements scaled precisely to the field dimensions declared by the user.

Research Contribution

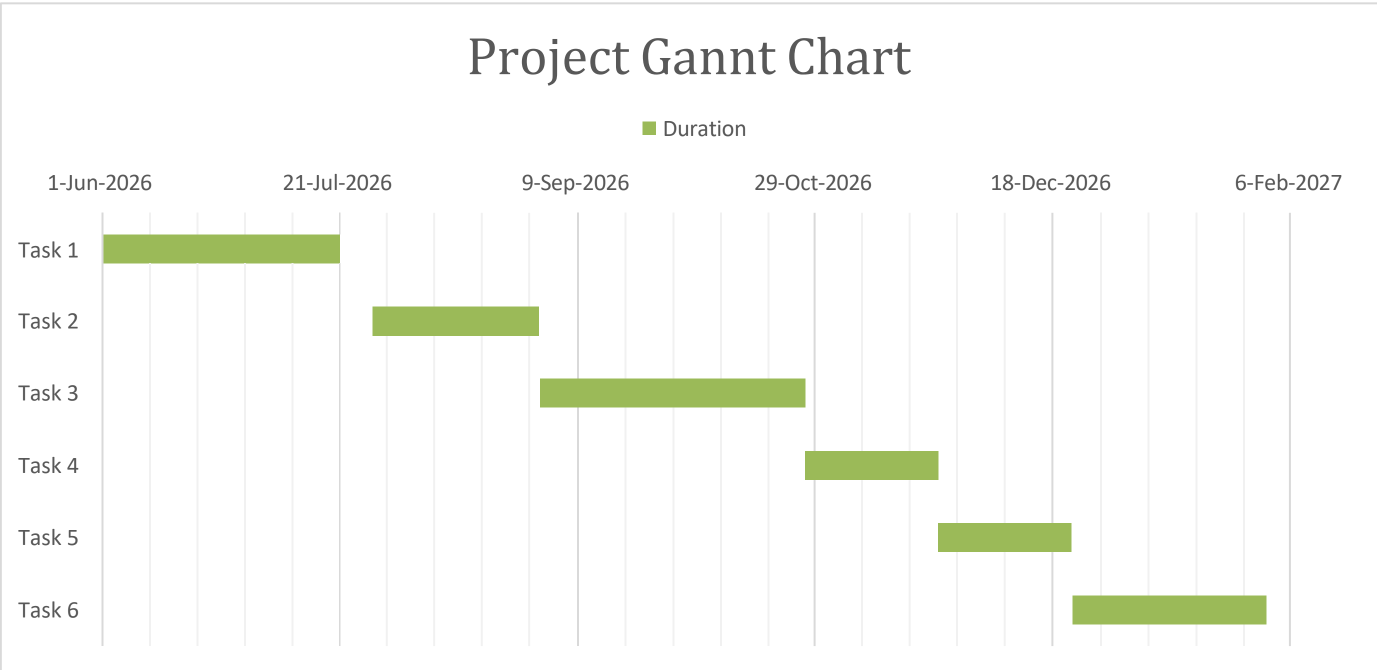
This study delivers key architectural enhancements to the domain of digital precision agriculture in developing South Asian regions. It introduces a new approach to disease classification: from static, one-label disease to complex, multi-label disease, and localizes the nutrient deficiencies, creating an easy-to-use framework for real-time monitoring of plant stress.

Socio-economic value: Embedding a multi-layered agronomic decision pipeline into a simple and easy-to-use mobile app makes scientific knowledge more accessible to smallholder farmers who are not specially trained in agronomy. In practice, this precision field-scaled approach helps reduce the risk of significant chemical over-application errors, increases the longevity of the soil profile and offers a sustainable solution to ensure long-term food security in Sri Lankan cultivation areas.

Project Timeline

The project spans an 8-month development cycle from June 2025 to January 2026:

Task No.	Research Task Description	Duration	Start Date	End Date
1	Requirement Gathering, Literature Review & Dataset Collection	8 Weeks	June 01, 2026	July 27, 2026
2	Development of Multi-Stage Image Preprocessing Pipeline	5 Weeks	July 28, 2026	August 31, 2026
3	YOLO Model Training, Tuning, and Severity Index Optimization	8 Weeks	Sept 01, 2026	Oct 26, 2026
4	Logic Architecture & Rule-Based Decision Matrix Formulation	4 Weeks	Oct 27, 2026	Nov 23, 2026
5	Mobile Application Packaging and Standalone API Integration	4 Weeks	Nov 24, 2026	Dec 21, 2026
6	Field Testing, Validation, and Thesis Report Compilation	6 Weeks	Dec 22, 2026	Jan 31, 2027



References

[1] S. &. H. A. (. Portch, "A systematic approach for balanced fertilizer recommendations for crops, Postgraduate Institute of Agriculture".

[2] K. W. C. K. U. M. S. &. S. S. (. Suresh, "How productive are rice farmers in Sri Lanka? The impact of resource accessibility, seed sources and varietal diversification".

[3] G. K. V. L. S. C. &. K. P. P. N. V. (. Udayananda, "Rice plant disease diagnosing using machine learning techniques: a comprehensive review. SN Applied Sciences,".

[4] D. O. C. A. S. S. LANKA, "PADDY STATISTICS 2023 YALA SEASON".

I hereby certify to the best of my knowledge that the details mentioned above are true and accurate.	
Signature of Candidate	**Date**
	28/05/2026
I/ We hereby certify to the best of my/ our knowledge that the details mentioned above are true and accurate.	
Signature of Supervisor (s)	**Date**
	29/05/2026
	29/05/2026